Birmingham geotechnical desk study

DRAFT D03 | 28 October 2026 | Edgbaston BH3 and TP1-TP6 revision | Public demonstration area

The additional Edgbaston records materially sharpen the follow-up priorities. TP4 records very loose ash and construction-debris fill above a bathing-pool base, pungent groundwater at 0.80 m and continuing excavation collapse. Locate and assess that historical feature before specifying GI where it may affect the works. The reviewed records are historical observations at archived positions; their relevance to the actual development footprint remains to be established.

Key findings from the originals

- TP4 reached a recorded bathing-pool base at 1.15 m beneath paving and very loose ash/cinders/brick/concrete fill. Pungent water and collapse are direct historical observations, requiring targeted history, environmental and excavation assessment. They do not identify a contaminant or a current groundwater regime. [SP08SE128 p1]
- BH3 records fill to 1.30 m and poorly cemented weak-to-moderately-weak sandstone from 6.40 to 8.10 m. Across BH1-BH3, local fill bases are 0.60-1.30 m and sandstone tops 5.50-6.50 m. Do not transfer these depths to the demonstration point. [SP08SE122/123/124 p1]
- TP1 shows contrasting materials at opposite pit ends and wet sand pockets; TP3 has possible fill and variable contacts; TP5 records buried topsoil; TP6 has a variable fill base. These support investigating short-distance changes and buried features. [SP08SE125/127/129/130 p1]
- BH3 is dated 6-7 September 1988 and TP1-TP3 8 September 1988, conflicting with index year 1989. The TP4-TP6 sheet has no excavation date or method; September 1988 is drawing/campaign context only. Earlier depth conflicts remain unresolved. [Originals and index follow-up]
- SPT seating blows over limited penetration remain separate from ordinary N values. Most Edgbaston holes have drilling/excavation-day dry notes; TP4 records water and TP1 wet pockets. No laboratory results or verified design parameters are supplied. [Edgbaston original sheets]

Revision and study basis

D03 supersedes D02. This revision visually reviews 28 pages in seven further BGS PDFs, adding BH3 and TP1-TP6. Together with the D02 review, 13 physical exploratory holes are represented by 13 PDFs and 40 archive pages. Repeated pit sheets and drawings are counted once as evidence. Source URLs, hashes, page references and uncertainties are retained in the audit JSON.

The frozen Groundwork run remains dated 28 October 2026, 19:30:45 UTC, for 52.45, -1.93, England, nominal radius 1 km: 11 queries, 13 HTTP 200 responses, one coverage entry, ten open tasks and seven unknown parameters. A separate bounded index follow-up retrieved 20 records. No site boundary, surveyed levels, works layout, loads or excavation geometry has been supplied; the app export is unchanged.

Working ground model and site relevance

This model combines regional mapping with separate historic observations. It identifies questions for the actual works; it is not a confirmed vertical profile at the query point. Depths below refer only to the individual archived holes.

Component	Observed evidence	Question for GI
Fill and buried features	Made ground in Bournbrook; fill in BH1-BH3; TP4 pool base beneath ash/debris; TP5 buried topsoil; possible fill at TP1/TP3.	Locate the former pool and other buried features; assess fill thickness, obstructions, settlement, water and environmental linkages.
Mixed soils	Clay, silt, sand and gravel vary between and within pits. TP1 contrasts north/south materials and wet pockets; BH1 includes very-soft-described clayey sand.	Investigate weak pockets, lenses and short-distance variability. Logged consistency/density words are historical descriptions.
Historical marl / rock bands	Bournbrook A: Keuper marl from 20 ft (6.10 m). Bristol A/B: marl from 8 ft 6 in / 6 ft (2.59 / 1.83 m), with deeper rock bands.	Resolve modern correlation, weathering, soil-rock transition and band hardness/thickness. No founding horizon or formation boundary is established.
Sand / weak sandstone	BH1/BH2/BH3 sandstone tops 6.50 / 5.50 / 6.40 m, continuing to 8.10 m. All record poor cementation, chiselling and limited-penetration blows.	Distinguish dense sand, weathered/cemented material and rock; obtain suitable samples/core where rock behaviour affects design.
Water and excavation behaviour	TP4: pungent water at 0.80 m and continuing collapse. TP1: wet pockets despite an overall dry/stable excavation note.	Resolve local water sources and current stability; select investigation and temporary-works methods accordingly.

Regional mapped context

BGS 1:625,000 queries returned Triassic sandstone/conglomerate and mudstone/siltstone/sandstone descriptions, plus glacial sand/gravel and till/diamicton. Each query returned two of two matches, with geometry omitted. These national-scale results do not identify the exact-point unit or its thickness. Formation-equivalent fields were blank. Release metadata are bedrock 11-2-2008, version 5.17, and superficial 30-4-2003, version 1.10, not established survey dates. [query-001/002; G1]

Obtain suitably detailed mapping, georeference the historic plans and compare geological setting, elevation, location precision and engineering influence depth with the actual footprint. Do not connect the southern marl and northern sandstone records into one asserted section or transfer their contact depths to the proposed site. Current conditions may differ after decades of development.

Original-log register and discrepancies

Original PDF	Date on original	Index depth / distance	Original depth evidence
SP08SE14	1960	4.00 m / 670.5 m	14 ft 6 in = 4.42 m; index differs.
SP08SE16	1959	15.00 m / 487.5 m	50 ft = 15.24 m; metric annotation/index 15 m.
SP08SE49	1963	18.29 m / 395.5 m	Table 60 ft = 18.29 m; note including pit 58 ft 6 in = 17.83 m.
SP08SE50	20 Oct-12 Nov 1962	15.24 m / 439.0 m	Table 50 ft = 15.24 m; note including pit 49 ft = 14.94 m.
SP08SE122	31 Aug-1 Sep 1988	8.10 m / 693.5 m	Log 8.10 m; index year 1989 differs.
SP08SE123	2-5 Sep 1988	8.10 m / 662.4 m	Log 8.10 m; index year 1989 differs.

All entries cite original PDF page 1. Imperial conversions use 1 ft = 0.3048 m, rounded here to 0.01 m. Original imperial entries take precedence for the stated conversion, but conflicting source statements remain unresolved. Tabulated depths describe the original log reference, with no verified present-day ground datum; Bristol pit/depth conventions require clarification.

Quality and location checks still required

The index gives all six a position precision of plus/minus 10 m. Distances are approximate straight-line distances to the query point, not the actual site. Four index drilling years were blank; the originals supply dates above. January 2013 timestamps are index updates. Bournbrook A records 401.5 ft O.D. with a later 122.35 m annotation; exact conversion is 122.3772 m. Do not use either as a surveyed site datum. The Edgbaston page-1 ground-level fields are blank.

Seek supporting reports to resolve total-depth conventions, sample records and the Edgbaston date discrepancy. Preserve the original index values in the frozen export and record the correction/conflict separately. Terminal depths alone do not establish the reason for termination, bedrock competence or an adequate design influence depth.

Archive coverage remains incomplete

The provider reported 423 SOBI matches in the enclosing rectangle; offsets 0 and 3 returned six, with continuation at offset 6. The six are not established as the nearest or most relevant records. The 423 is not an exact-circle count, a count of distinct physical holes or a guarantee of exhaustive archive coverage. Review of these six originals closes their acquisition/review step only. [query-003/004]

The separate D03 index follow-up used the same box, offset 6, limit 20 and returned 20 more features: 26 distinct index candidates now obtained across the two retrieval stages, while 13 requested holes have original records reviewed. The provider still reports 423 matches and continuation at offset 26. This remains a partial search. [index-follow-up]

The BH3 and TP1-TP6 acquisition/review gap is now closed for the retrieved PDFs. The complete narrative report, laboratory results, historic survey basis, current site relevance and wider archive coverage remain unresolved. An empty AGS result does not establish absence of usable files. [query-005; G2]

Older southern logs: recorded strata

Summaries below paraphrase the original page-1 logs. Metric limits are conversions of the imperial tables, not new measurements. Detailed interval transcriptions and precision are retained in the accompanying log-review JSON.

Record	Original depth interval	Recorded materials / observation
SP08SE14 1960	0-0 ft 9 in (0-0.23 m)	Topsoil.
SP08SE14	0 ft 9 in-7 ft (0.23-2.13 m)	Several sandy/gravelly and silty clay layers; brick traces in the upper layer. Firm clay locally.
SP08SE14	7 ft-8 ft 9 in (2.13-2.67 m)	Clayey sand with clay pockets, followed by sand/gravel with clay traces.
SP08SE14	8 ft 9 in-14 ft 6 in (2.67-4.42 m)	Brown friable silty clay, grey clayey silt, then very friable brown clayey silt.
SP08SE16 1959	0-11 ft 6 in (0-3.51 m)	Made ground.
SP08SE16	11 ft 6 in-20 ft (3.51-6.10 m)	Blackish-brown clay with sand/gravel traces.
SP08SE16	20-50 ft (6.10-15.24 m)	Historical Keuper marl description; later metric end annotation differs.
SP08SE49 1963	0-8 ft 6 in (0-2.59 m)	Brick/mortar/timber/stones to 2 ft (0.61 m), then mixed clay with brick, stone and earthenware.
SP08SE49	8 ft 6 in-18 ft 6 in (2.59-5.64 m)	Friable brown/grey marls.
SP08SE49	18 ft 6 in-60 ft (5.64-18.29 m)	Friable brown/grey marls with rock bands; table total conflicts with separate pit-inclusive note.
SP08SE50 1962	0-6 ft (0-1.83 m)	Mixed clay with brick/stones to 3 ft (0.91 m), then silty clay, pebbles, sandy pockets, brick and roots.
SP08SE50	6 ft-14 ft 3 in (1.83-4.34 m)	Friable brown/grey marls.
SP08SE50	14 ft 3 in-38 ft (4.34-11.58 m)	Friable brown/grey marls with rock bands.
SP08SE50	38-50 ft (11.58-15.24 m)	Friable brown/grey marls with very hard rock bands; conflicting total-depth note.

No drilling method is established confidently for these four sheets. SP08SE14/16/49 show no groundwater or test results; blank fields are not dry observations. SP08SE50 records a pit, tubes withdrawn on completion and ten nominally undisturbed samples, but no laboratory results. Historical marl labels and hardness words do not establish modern formation identity, intact strength or bearing resistance. [SP08SE14/16/49/50 p1]

Edgbaston logs: two local profiles

SP08SE122 is original BH1; SP08SE123 is BH2 at the Lucas Institute Annexe, University of Birmingham. Both used 150 mm cable percussion, with casing to 8.0 m, and end at 8.10 m. The original dates are in 1988. The log sheets give no explicit termination reason or ground elevation. [Both p1]

SP08SE122 / BH1: m below logged ground	SP08SE123 / BH2: m below logged ground
0-0.20: sandy topsoil labelled fill. 0.20-1.10: soft silty sandy clay fill with gravel, occasional cobbles and brick.	0-0.20: sandy topsoil labelled fill. 0.20-0.60: soft-firm silty sandy clay fill with brick, gravel and occasional cobbles.
1.10-1.30: very-soft-described clayey sand. 1.30-2.30: firm sandy clay with gravel.	0.60-1.60: firm, becoming stiff-very stiff, sandy clay with gravel.
2.30-4.00: dense gravelly coarse sand, becoming coarse sandy gravel.	1.60-4.65: dense-very dense coarse sandy gravel, with clayey-sand pockets towards base.
4.00-4.30: firm-stiff sandy clay.	No directly corresponding separate clay layer is logged; do not infer continuity from BH1.
4.30-6.50: medium-dense fine sand with occasional sandstone lithorelicts.	4.65-5.50: medium-dense fine sand, becoming very dense.
6.50-8.10: very poorly cemented fine sandstone, weak, becoming moderately weak.	5.50-8.10: very poorly cemented fine sandstone, very weak, becoming weak-moderately weak.

All intervals above: original PDF page 1; material and consistency/density/strength descriptions are historical log observations. The parallel columns do not assert cross-hole correlation. Fill bases differ by 0.50 m and sandstone tops by 1.00 m at these two holes; neither difference can be transferred to the demonstration point.

Shared drawings and further records

The three borehole PDFs contain individual logs at page 1. The six trial-pit PDFs repeat two combined sheets: TP1-TP3 and TP4-TP6. All nine Edgbaston PDFs repeat the same September 1988 drawings at pages 2-4: location plans, exploratory-hole plan and geological section (contract 1829, drawings 1-3). The plan shows the nine physical holes, a historic proposed-building outline, retaining wall and vegetation; it is not a current survey.

The west-east section uses questioned correlations and extrapolated ground levels. Its horizontal 1:200 and vertical 1:100 scales imply twice vertical exaggeration; no surveyed section alignment is established. The interpreted boulder-clay key does not replace individual descriptions, and its groundwater key is not a measurement. Use original log sheets for test entries and obtain the full report/survey basis before adopting correlations. [Edgbaston pp2-4]

Additional records and Edgbaston BH3

Original PDF	Hole	Index / log depth m	Original date / status
SP08SE124	BH3	8.10 / 8.10	6-7 Sep 1988; index 1989 differs.
SP08SE125	TP1	1.80 / 1.80	8 Sep 1988; index 1989 differs.
SP08SE126	TP2	2.40 / 2.40	8 Sep 1988; index 1989 differs.
SP08SE127	TP3	2.00 / 2.00	8 Sep 1988; index 1989 differs.
SP08SE128	TP4	1.15 / 1.15	Excavation date absent; Sept 1988 drawings. Index 1989 unverified.
SP08SE129	TP5	2.00 / 2.00	Same undated combined sheet as TP4.
SP08SE130	TP6	2.00 / 1.80	Depth conflict; undated combined sheet. Index 1989 unverified.

All log observations cite page 1 of the linked PDF. The index has plus/minus 10 m position precision. TP6 ends at 1.80 m while the index says 2.00 m; this discrepancy is unresolved. Retain the 1989 index year separately from explicit/missing original dates. TP1-TP3 state JCB 3C excavation; their date/method cannot automatically be assigned to TP4-TP6.

BH3 profile: depth below logged ground, metres

Interval	Paraphrased original observation
0-0.20	Sandy topsoil labelled fill.
0.20-1.30	Soft-firm sandy clay fill, sand pockets and occasional gravel.
1.30-3.90	Very dense sand/gravel, becoming dense.
3.90-4.30	Stiff sandy clay with gravel.
4.30-5.00	Medium-dense fine sand.
5.00-5.30	Clayey fine sand, described as very stiff.
5.30-6.40	Dense fine sand, becoming very dense.
6.40-8.10	Very poorly cemented fine sandstone, weak-moderately weak.

BH3 used 150 mm cable percussion, casing to 8.0 m. Ground-level field and explicit termination reason are absent. Water was added below 1.30 m; drilling-day dry notes occur on 6 and 7 September. One hour chiselling at 6.40-8.00 m supports allowing for cemented material in method selection, without establishing competent bearing rock. [SP08SE124 p1]

Trial pits: findings and implications

These are six physical pits recorded on two combined sheets, reproduced in six BGS PDF packages. Depths are interpreted as metres from the record series, section and index. Slash boundaries denote variable contacts within a pit; no midpoint has been substituted. TP1-TP3 are explicitly dated 8 September 1988; TP4-TP6 dates are unrecorded on their sheet.

Pit / depth	Paraphrased ground observations	Water, stability and sample notes
TP1 1.80 m	Topsoil 0-0.30; soft-described, slightly clayey fine sand 0.30-0.60. At 0.60-1.50, south end firm-stiff silty clay; north end possible gravelly fill with wet sand pockets. Gravel/cobbles 1.50-1.80.	Overall dry/stable during excavation, despite local wet pockets. D sample 1.00 m.
TP2 2.40 m	Topsoil to 0.30/0.40 with brick traces; sandy clay to 0.80; fissured stiff-very stiff silty clay to 2.15; fine sand with soft clay lumps to 2.40.	Dry/stable during excavation. No sample/test result recorded.
TP3 2.00 m	Topsoil to 0.10/0.25; possible fill of soft-firm very clayey sand/gravel to 0.70/0.90; very stiff silty clay to 1.80; sandy gravel to 2.00.	Dry/stable during excavation. No sample/test result recorded.
TP4 1.15 m	Paving slabs on fine sand to 0.15; very loose ash/cinders/brick/concrete fill to 1.15; bathing-pool base recorded at 1.15. Natural ground below is not proved.	Pungent groundwater, level 0.80 m; continuing collapse. D sample 1.00 m; no analytical result.
TP5 2.00 m	Fill topsoil to 0.20; soft sandy-clay fill with tile to 0.70; buried topsoil 0.70-1.00; firm-very stiff silty clay to 1.80; gravel to 2.00.	Dry/stable during excavation. No sample/test result recorded.
TP6 1.80 m	Fill topsoil to 0.30; soft clayey-sand fill with tile/topsoil pockets to 0.60/0.70; soft-firm clay to 1.10; stiff clay to 1.50; gravel/cobbles to 1.80.	Dry/stable during excavation. No sample/test result recorded.

Sources: TP1-TP3, SP08SE125/126/127 p1; TP4-TP6, SP08SE128/129/130 p1. D is conventionally interpreted as disturbed; the pit sheets do not define the code. No analytical results or explicit termination reasons are given; TP4 ends at the recorded pool base. Excavation dimensions, observation times and equilibrium water levels are absent.

Change to priorities

First locate the former pool and assess its extent, infill, water source and relationship to proposed works. Pungent water plus ash/debris justifies a competent-person environmental conceptual model and evidence-led sampling; no chemical identity or contamination concentration is established. Historical collapse requires a reviewed investigation/temporary-works approach. Elsewhere target fill changes, buried topsoil, wet pockets and granular/clay contacts rather than assuming a uniform shallow profile.

Testing and groundwater: usable limits

Historic test information

Unstarred SPT-column entries range 4-77 in BH1, 14-141 in BH2 and 12-85 in BH3; 141 at 5.50 m is printed on BH2. Energy/overburden corrections, complete test sheets and a verified modern interpretation basis are absent. No strength/density correlation is adopted. The legend defines starred entries below as seating blows, not ordinary full-penetration N values.

Record / depth	Restricted-penetration entries	Interpretation limit
BH1 6.50, 7.00, 7.50, 8.00 m	94* /150 mm; 53* /75 mm; 51* /75 mm; 64* /75 mm, respectively.	The legend defines * as SPT seating blows. These entries are not ordinary full-penetration N values.
BH2 6.00, 6.50, 7.00, 7.50, 8.00 m	52*, 61*, 60*, 65*, 70*, each /75 mm, respectively.	Keep blows and penetration together. Do not pool with unstarred entries or infer competent rock from high blows.
BH3 6.50, 7.00, 7.50, 8.00 m	53*, 66*, 64*, 72*, each /75 mm, respectively.	Same seating-blow qualification; use original logs rather than section duplicates.

One hour chiselling is recorded in each borehole: BH1 6.50-8.00 m, BH2 5.90-8.00 m, BH3 6.40-8.00 m. No core recovery, RQD or rock-strength results are supplied. Bristol B reports ten nominally undisturbed samples without depth schedules/analyses; TP1 and TP4 note D samples at 1.00 m without analytical results. [Original log sheets]

Water observations and their limitations

- Bristol Road B: standing-water notation 16 ft 7 in (about 5.05 m). Reference datum, measurement date/time and stabilisation are not established; do not report it as a verified depth below present ground or an AOD elevation. [SP08SE50 p1]
- Edgbaston BH1/BH2/BH3: drilling-day dry notes, with added water below 2.30/1.60/1.30 m respectively. No monitoring installation or measured equilibrium level is recorded. These do not establish absence of groundwater. [SP08SE122/123/124 p1]
- TP4: pungent groundwater, recorded level 0.80, interpreted nominally in metres in the pit-depth convention; the sheet gives no explicit datum, time, stabilisation or surveyed elevation. TP1 has wet pockets despite its overall dry/stable note. Other pits give excavation-day observations only. [Pit sheets p1]
- Nearby EA Birmingham University station: latest returned value 129.14 mAOD on 16 June 2026 at 09:28:28, quality Unchecked, 134 calendar days before retrieval. The station is approximately 200 m from the point. A Qualified series label does not override reading quality. [query-009/011; G3]

mAOD is elevation, not depth. Station datum=135.23 is not a verified site ground level. Assess aquifer relevance, perched water, pore pressures and seasonal variability for the works. Do not combine the 1962 notation, historic borehole/pit observations and 2026 station elevation into a single groundwater profile.

Remaining gaps and closure criteria

Priorities indicate sequence and prerequisites, not assessed hazard likelihood. Review of the seven requested additional PDFs is complete; together with D02, thirteen physical holes have original-record review. The frozen app counts are unchanged. Close remaining work packages only after recorded technical review.

Priority	Gap after original-log review	Required closure evidence
P0	Site and engineering scope	Boundary, surveyed levels, proposed structures, loads, excavations, drainage and movement criteria.
P0 gate	Utilities, access and permissions	Asset-owner information, suitable service locating/survey, access and an agreed safe intrusive method before fieldwork. [G4]
P1 first	TP4 former pool, ash fill, water and collapse	Locate/georeference feature; assess history, extent, current water and environmental linkages; agree investigation and temporary-works controls where it affects the works.
P1	Record relevance and completeness	Georeferenced historic plans; wider archive search; full Edgbaston narrative/lab report and survey basis; Bristol sample/lab records; TP6 and earlier discrepancy decisions.
P1	Site history / ground hazards	Detailed mapping, land-use and development history, walkover, relevant mining/ground-hazard information and environmental conceptual model.
P1-P2	Fill and mixed-soil variability	Footprint-specific evidence of fill base, obstructions, soft pockets and granular/clay lenses, with suitable sampling and classification.
P2	Weathered material / rock bands	Applicable depth coverage, weathering, sand-sandstone distinction and band/core quality where design requires; verified test quality.
P1-P2	Groundwater regime	Relevant station history and project-specific monitored levels/pore pressures where needed, with surveyed datum, response zones and justified observation coverage.
P2-P3	Engineering parameters	Quality-controlled field/lab evidence addressing actual works and a reasoned interpretation of applicability, uncertainty and selected values.

Other queries do not close these gaps

Empty AGS, landslide, EA flood-monitoring-station and current-warning results establish only that those requests returned no records. They do not prove absence of investigations, instability, flood hazard or groundwater. Review long-term flood/drainage information for the works. [query-005/008; G6]

Planning Data identified Birmingham LPA, entity 626130, reference E60000130; its representative point is not a boundary. TP4 ash/debris and pungent water warrant a competent-person environmental assessment; no contaminant or concentration is established. Mining effects, shrink-swell, faults and other relevant hazards require targeted review. [query-010; G5/G7]

Prioritised plan: scope and desk work

P0 Confirm the works and decision needs

Client and geotechnical designer: establish the site polygon, topographic survey, existing and proposed levels, loads, serviceability criteria, earthworks, drainage and any retained excavations. Define which foundation, settlement, excavation and groundwater decisions the GI must support. Required output: controlled site/works plan, investigation brief and constraints register.

P1 Resolve evidence relevance before final GI design

- First georeference all thirteen reviewed hole positions and historic plans against the actual footprint. Prioritise locating the TP4 bathing pool and its relationship to works, drainage and neighbouring assets. Verify current condition and infill history; the pit proves neither the full pool extent nor natural ground below its base.
- Obtain the full Edgbaston narrative, laboratory schedules/results and survey/drawing datum. BH3 and TP1-TP6 log retrieval is now complete. Seek Bristol supporting records and resolve the TP6 2.00 m index versus 1.80 m original depth, earlier total-depth conflicts and original/index dates where possible.
- Extend relevant archive coverage beyond the 26 distinct index candidates now obtained; the provider reports 423 box matches. Use bounded requests or publisher/GIS tools and record unsearched coverage. Search for available AGS and client-held records.
- Obtain detailed geological and relevant ground-hazard information, historic maps and land-use/development records. Walk over the site to check made-ground indicators, obstructions, retaining walls, drainage, slopes, vegetation, neighbouring assets and access.
- Coordinate competent-person environmental assessment. Give TP4 ash/cinders, pungent water and the former pool explicit source-pathway-receptor questions; develop chemical, water or gas sampling only from that model and site evidence. Review legacy sample results if obtainable. [G5]

Required output: a site-specific conceptual model, record-relevance register and preliminary GI scope. Include TP1 lateral changes, TP3 possible fill, TP5 buried topsoil and TP6 variable contacts. Log retrieval is complete for the thirteen reviewed holes; full-report, laboratory and relevance gaps remain.

Hold point before intrusive work

Obtain utility information and suitable service locating/survey; confirm permissions, working space, access and health/environmental controls. Where the former pool or loose fill may affect investigation, explicitly review collapse, buried structures and water; select suitable plant and temporary works or alternative methods. Agree a safe intrusive method. HSG47 addresses planning, service locating and safe excavation. Public records provide no utility clearance. [G4]

Release criterion: designer has reviewed the works, conceptual model and investigation objectives, and responsible parties have agreed the scope and constraint controls. This demonstration point does not justify fixed hole quantities, spacing, depths, a foundation type or a GI price.

Targeted GI, monitoring and interpretation

P2 methods below are conditional options to resolve the evidenced questions. The GI designer should select locations, quantities and depth coverage from the actual footprint, loading and engineering influence zone, and update the scope as findings emerge.

Priority target	Field and laboratory response	Decision supported
Former pool / ash fill / water	After locating TP4 and reviewing the environmental model, target fill/pool margins and water where relevant. Use methods addressing collapse and buried obstructions; obtain representative samples and evidence-led analyses.	Pool/infill extent, current water source, environmental linkages and safe investigation/excavation response.
Fill, obstructions and soft pockets	Use safe shallow trial pits or suitable sampling where feasible. Sample variable fill and natural soils; classify grading, moisture and plasticity, with compressibility/strength work where relevant.	Fill extent/base, earthworks reuse, variability and settlement/excavation questions.
Clay / sand / gravel transitions	Boreholes and suitable sampling/testing across relevant strata. Select penetration or other in-situ testing with a verified quality basis; account for gravel/cobbles and local clay lenses.	Continuity, sample quality and project-specific strength, stiffness and permeability interpretation.
Weathered marl / rock bands / sandstone	Choose plant capable of advancing through hard bands and weak rock. Consider rotary coring where rock behaviour governs, with recovery/discontinuity/weathering records and appropriate tests.	Distinguish weathered soil, cemented material and rock; assess rock quality and excavation/foundation response.
Groundwater and environmental questions	Select properly sealed monitoring installations and response zones after strata review; survey elevations and obtain repeat observations. Coordinate chemistry/gas work where the environmental model justifies it.	Relevant water levels and pore pressures, variability, excavation/drainage questions and evidenced environmental linkages.

Testing and monitoring scope

Schedule laboratory strength, stiffness/compressibility, permeability and chemical tests only where materials and proposed works require them, with sample quality and relevant stress/groundwater conditions specified. Historical SPT entries and seating blows cannot supply design correlations without a justified applicability and quality review. Retrieve old laboratory results where possible; new data may still be needed.

Review station/aquifer history and the possibility of local perched water. Design monitoring to address temporal variability relevant to the works, with recorded installation details, datum and quality. The current evidence does not justify a fixed monitoring duration, seasonal maximum, installation depth or dewatering rate.

P3 Interpret, review and close

Check new factual logs, sample/test quality, locations, levels and monitoring results. Reconcile the ground model with the proposed works, identify residual uncertainty and commission supplementary investigation where needed. Record the applicability and uncertainty of derived inputs and the technical reviewer decision.

Design release requires evidence addressing critical engineering questions. Friction angle, undrained strength, stiffness, permeability, design groundwater level, allowable bearing pressure and utility clearance remain unknown in D03. Workflow closure does not establish a parameter; project-specific technical review remains required.

Sources, attribution and audit trail

D03 sources: seven additional BGS PDFs retrieved 28 October 2026 at 21:17:06-21:17:18 UTC, all 28 pages visually reviewed. D02 supplied the earlier six PDFs/12 pages. Combined coverage is 13 physical holes, 40 archive pages and repeated sheets counted once. The audit JSON records hashes, dates, page references, uncertainty and the separate 20-record index follow-up. Original scans, drawings and personal details are not redistributed.

Derived review: examples/birmingham_original_log_review_2026-10-28.json. The frozen Groundwork evidence remains examples/birmingham_report_evidence_2026-10-28.json. It retains query arguments, response dates/hashes, metadata, warnings, pagination and source terms. The following links are each query's first provider request; additional requests remain in that evidence file.

Query	Evidence	Status	Returned	Provider request
query-001	Regional geology	ok	2	Request
query-002	Regional geology	ok	2	Request
query-003	Historic index	partial	3	Request
query-004	Historic index	partial	3	Request
query-005	Historic index	empty	0	Request
query-006	Landslide index	empty	0	Request
query-007	EA stations	empty	0	Request
query-008	Current warnings	empty	0	Request
query-009	Groundwater stations	ok	1	Request
query-010	Planning authority	ok	1	Request
query-011	Groundwater reading	ok	1	Request

Attribution and terms

Contains British Geological Survey materials (c) UKRI 2026. The open SOBI index and 625K mapping have their stated OGL scope; linked scans/reports can carry separate underlying rights. Original records were consulted for factual summaries; access is not a grant to redistribute their contents. Landslide inventory terms include an insurance-use restriction. Contains Environment Agency information licensed under OGL v3.0.

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[G1] BGS Geology 625K. Accessed 28 October 2026.

[G2] BGS Single Onshore Borehole Index. Accessed 28 October 2026.

[G3] EA Hydrology API reference. Accessed 28 October 2026.

[G4] HSE HSG47 Avoiding danger from underground services. Accessed 28 October 2026.

[G5] EA Land Contamination Risk Management Stage 1. Accessed 28 October 2026.

[G6] GOV UK long term flood risk service. Accessed 28 October 2026.

[G7] Planning Data API documentation. Accessed 28 October 2026.

Index follow-up evidence: examples/birmingham_edgbaston_index_followup_2026-10-28.json. It preserves the MCP-adapter query, 20 returned features, source terms, request time/hash and remaining pagination. This extends discovery without changing the frozen app export. [index-follow-up]

Frozen Groundwork evidence SHA-256: a8ccd4e9dd99e1e7d5c32c5b895cef50448f1be28273df6102aa45719f39f2b6. D03 is a draft for project scoping and technical review.